

DFT from scratch

Cheat sheet · placeholder document

- **Kohn–Sham equation**

$$[-\frac{1}{2}\nabla^2 + v_{\text{eff}}] \psi_i = \epsilon_i \psi_i$$

- **Density**

$$n(r) = \sum_i |\psi_i(r)|^2$$

- **Effective potential**

$$v_{\text{eff}} = v_{\text{ext}} + v_{\text{H}} + v_{\text{xc}}$$

- **Hartree (Poisson)**

$$\nabla^2 v_{\text{H}} = -4\pi n$$

- **Self-consistency**

mix densities until n converges